Water Research 279 (2025) 123438

Contents lists available at ScienceDirect

Water Research

journal homepage: www.elsevier.com/locate/watres

DFT-assisted machine learning for polyester membrane design in textile wastewater recovery applications

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ARTICLE INFO

Keywords:
Machine learning
Polyester membrane, Density functional theory
Resource recovery
Textile wastewater

ABSTRACT

Resource recovery from textile wastewater has attracted increasing interest because it simultaneously addresses wastewater treatment and maximizes the utilization of the residual dyes. Although polyester membranes have demonstrated great potential for textile wastewater recovery, tailoring high-performance polyester membranes remains a multidimensional challenge because of the complex nonlinear relationships between the membrane materials and their performance. Here we developed density functional theory (DFT)-assisted machine learning models that integrates DFT descriptors with fabrication and operation parameters to facilitate the generative design of polyester membranes. The developed machine learning model demonstrated the ability to accurately predict permeance and separation performance. The contribution analysis revealed that the fabrication parameters emerged as the critical factors influencing permeance, whereas the DFT descriptors played important roles in determining the dye and salt rejection. Additionally, optimal combinations of monomer, fabrication, and operation conditions were identified from a chemical space of 8,000 candidates using the developed model combined with Bayesian optimization, targeting dye/salt and dye/dye selectivity. Five polyester membranes were then fabricated under these identified combinations. These membranes surpassed the current performance upper bound and achieved efficient recovery of the dyes from textile wastewater. Overall, a feasible and universal machine learning model aimed at driving a paradigm shift in the inverse design of polyester membranes was developed.

1. Introduction

The textile industry is the second largest source of pollution and accounts for approximately 20 % of global water pollution (Grant et al., 2012). As the dyeing process relies primarily on weak physical or chemical adsorption, approximately 200,000 tons of dyes are wasted and subsequently discharged in textile wastewater annually, posing substantial risks to ecosystems and human health (Al-Tohamy et al. 2022). Since the dye market is expected to grow to USD 10.13 billion in 2026, resource recovery from textile wastewater not only addresses the treatment challenges but also complies with sustainable development (Uddin 2021). In addition, estimates indicate that the production of 1 kg of the dye uses of up to 1.5 kg of salts and 40 kg of water (Hassaan et al. 2017). Traditional textile wastewater treatment technologies, such as adsorption, chemical precipitation and advanced oxidation, can effectively reduce the pollutant concentrations; however, they fail to achieve recovery (Fu et al. 2024; Li et al. 2022b; Periyasamy 2024). Therefore, the development and application of efficient textile wastewater recovery technologies are essential for the sustainable advancement of the textile industry.

Pressure-driven membrane technology provides a promising approach for the dye/salt and dye/dye selectivity in textile wastewater treatment by rejecting macromolecule dyes larger than the pore size and allowing micromolecule dyes or salts to pass through. (Jia et al. 2023; Mauter et al. 2018; Yasmeen et al. 2023). In this context, polyester membranes have gained increasing attention for textile wastewater

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https://doi.org/10.1016/j.watres.2025.123438
Received 20 December 2024; Received in revised form 11 February 2025; Accepted 3 March 2025
Available online 5 March 2025
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recovery due to their tunable membrane pore sizes, ranging from the nanoscale to the angstrom level, which can be attributed to the relatively low reactivity of polyhydric alcohols or polyphenols with 1,3,5-benzenetricarbonyl chloride (Cheng et al. 2022; Jin et al. 2022; Jin et al. 2021). Zhao et al. fabricated a polyester membrane with a molecular weight cut-off of 5839 Da using quercetin, this membrane significantly improved the dye/salt selectivity efficiency by 288 % with respect to that of a commercial NF270 nanofiltration membrane (Zhao et al. 2022a). However, current polyester membranes still suffer from a trade-off effect between the permeance and selectivity due to the nonoptimal combination of the monomers, fabrication and operation parameters, resulting in performance that falls below the theoretical upper bound (Li et al. 2024) . Moreover, screening for the optimal combination remains a multidimensional challenge because of the complex nonlinear relationships between the membrane fabrication process and performance. For example, by simply increasing the concentration of the aqueous monomer, a polyester membrane fabricated with glucose for dye/salt selectivity could be modified to one suitable for the separation of divalent salts (Jin et al. 2022; Zheng et al. 2021). Therefore, identifying optimal combinations from multidimensional availability is essential for achieving performance that exceeds the theoretical upper bound in polyester membranes.

Machine learning can recognize complex patterns, correlations, and trends within datasets and is increasingly being adopted in the field of membrane research; type of research includes prediction of water permeance, ion/ion selectivity, and emerging pollutant rejection based on membrane properties (Gao et al. 2023; Tao et al. 2023; Wang et al. 2024). However, few studies have integrated potential monomers with machine learning models to directly predict membrane performance. This is attributed to the difficulty in ensuring that the molecular descriptors used successfully capture the comprehensive information on the potential monomers, and these descriptors largely determine the predictive accuracy of the machine learning models (Gao et al. 2023; Sutariya et al. 2025). Molecular descriptors, such as mathematical representations of chemical compounds, can be derived from quantum principles with more detailed insights and include those provided by density functional theory (DFT). Integrating these insights with machine learning models has the potential to substantially expedite progress in material science, thereby reducing the material development cycle and minimizing research costs (Aldossary et al. 2024; Huang et al. 2023). For example, Fang et al. employed a DFT-assisted machine learning model to expedite the discovery of dual-metal-site catalysts with high electrocatalytic performance for renewable energy applications (Fang et al. 2024). Therefore, the development of such a model is expected to accelerate the tailoring of polyester membranes for textile wastewater recovery applications. However, a comprehensive investigation into its application for high-performance membrane design is lacking.

In this study, DFT-assisted machine learning models were developed and integrated DFT descriptors, fabrication and operation parameters to accurately predict the permeance and selectivity of the polyester membranes. Both the global and local contributions of the input features were quantified using Shapley additive explanation (SHAP) analysis and classification mapping. Potential aqueous monomers and their corresponding fabrication processes were identified via Bayesian optimization from a chemical space of 8000 candidates, which target the high dye/salt or dye/dye selectivity of polyester membranes. Five polyester membranes were experimentally fabricated using the identified parameters to evaluate their performance in textile wastewater recovery.

2. Materials and methods

2.1. Dataset construction and analysis

The workflow of machine learning for the improvement in the development of polyester membranes is outlined in Fig. 1. The input features were divided into three categories: DFT descriptors, fabrication parameters and operation parameters. To minimize the prediction errors arising from the limited sample size of organic monomers, 1,3,5-benzenetricarbonyl chloride was only selected as the organic monomer. Thus, the DFT descriptors mainly represent the properties of aqueous monomers and dyes. In addition, eight fabrication parameters, such as the aqueous monomer concentration, deacid reagent concentration and organic monomer concentration, were used as input features. The operation parameters varied depending on the prediction targets of permeance, dye rejection, and salt rejection. Sixteen dyes were used to evaluate the performance of the polyester membranes (Table S1). Briefly, the DFT descriptors of dyes were added to the dye rejection dataset as operation parameters, whereas five essential properties of the salt were introduced into the salt rejection dataset. The detailed input features are presented in Text S1.

The relationships between the input and target variables were analysed using the Pearson correlation coefficient (R_{ab}) (Schober et al. 2018), which was calculated via the Pandas package and expressed by Eq. (1):

$$R_{ab} = \frac{\sum_{i=1}^{N}(a_i - a^{ave}) - \sum_{i=1}^{N}(b_i - b^{ave})}{\sqrt{\sum_{i=1}^{N}(a_i - a^{ave})^2 - \sum_{i=1}^{N}(b_i - b^{ave})^2}} \quad (1)$$

where a_i and b_i denote the individual input and target variables, while a^{ave} and b^{ave} represent the respective means of variables, and the symbol

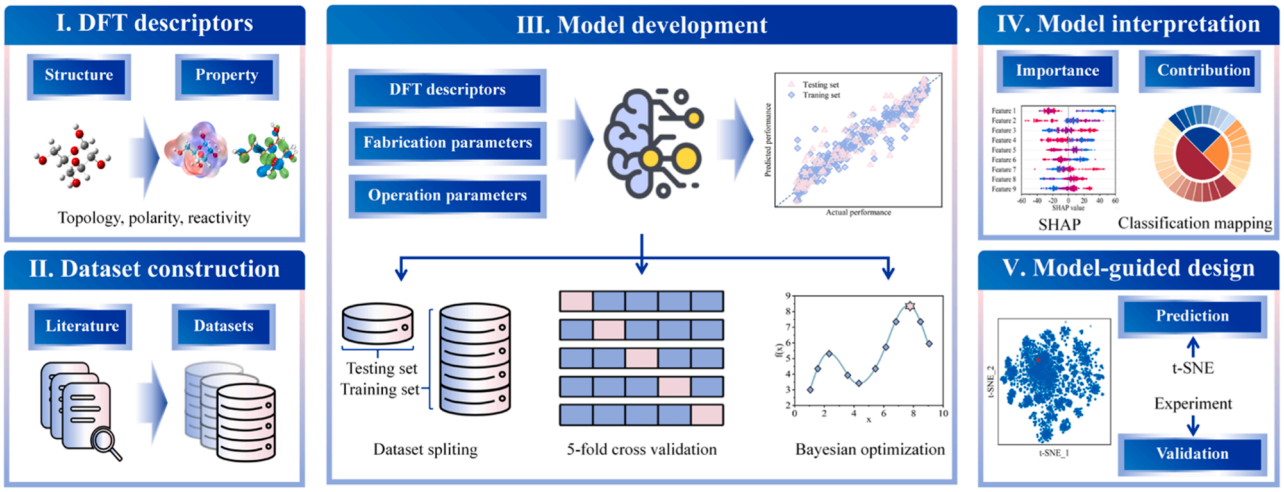


Fig. 1. Workflow of DFT-assisted machine learning algorithms for polyester membrane design.

N represents the total number of data points.

2.2. Density functional theory calculation of molecular properties

The features of the aqueous monomers and dye molecules were obtained from the DFT calculations using Gaussian 09W. The geometric optimization of molecules was performed using the B3LYP hybrid functional and 6–311G(d,p) basis sets to obtain minimum energy structures, followed by harmonic vibrational frequency calculations at the same level. Since the monomers and dye were dissolved in the aqueous solution, the solvation effect of the molecules was determined using the solvation model based on density. The single-point energy of each molecule was subsequently calculated at the B3LYP-D3/def2TZVP level. Finally, the DFT descriptors were obtained using the topological analysis, quantitative surface analysis and molecular orbital composition analysis via the Multiwfn 3.8 package (Lu 2024; Lu and Chen 2012). Based on the Pearson correlation and model results, sixteen DFT descriptors were selected and included the energy of the highest occupied molecular orbital (E_{HOMO}) and lowest unoccupied molecular orbital (E_{LUMO}), the maximum (Vs_{max}) and minimum (Vs_{min}) values of the electrostatic potential surfaces, the area of the molecular surface with a positive (Vs_+) and negative (Vs_-) electrostatic potentials, the average (AVs) and variance (RVs) of the overall electrostatic potential, the molecular polarity index (MPI) and the polar surface area (PSA). The specific definitions of the DFT descriptors are provided in Table S2.

2.3. Machine learning model development and evaluation

The extreme gradient boosting (XGBoost) model and convolutional neural network (CNN) model were selected to predict the performance of the polyester membranes (Fig. S1). The XGBoost model employs the gradient descent method to iteratively update parameters, which was widely used by data scientists (Huang et al. 2025; Zheng et al. 2025). By incorporating a regularization term into the loss function and employing unique techniques such as pre-stopping and post-pruning, the overfitting problem can be effectively mitigated while enhancing the convergence speed (Chen and Guestrin, 2016). The CNN model uses a convolutional architecture with features such as downsampling, weight sharing, and local connections to address challenges such as slow convergence and overfitting (LeCun et al. 2015). Before model development, the dataset was split into 80 % for the training set and 20 % for the testing set. The hyperparameters of the models were optimized using Bayesian optimization combined with 5-fold cross-validation (Tables S3–8). The prediction performance of the models was evaluated by the coefficient of determination (R^2), mean absolute error (*MAE*) and root mean squared error (*RMSE*), which are calculated using Eqs. (2)-(4):

$$R^2 = 1 - \frac{\sum_{i=1}^{N}\left(y_i - y_i^{pre}\right)^2}{\sum_{i=1}^{N}\left(y_i - y^{ave}\right)^2} \tag{2}$$

$$MAE = \frac{1}{N}\sum_{i=1}^{N}\left|y_i^{ave} - y_i\right| \tag{3}$$

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(y_i - y_i^{pre}\right)^2} \tag{4}$$

where y_i and y_i^{pre} represent the actual value and predicted value of sample *i.*, respectively, and y^{ave} is the average of y_i in the testing set.

2.4. Model interpretation

Model interpretation was analysed using SHAP, which is a cooperative game theory-based approach. The core concept of SHAP is to decompose the prediction result of the model into the contributions of individual features (Lundberg et al. 2020). In the SHAP analysis, the model was trained using all possible combinations of input features, and the differences between the predicted results for a specific input feature were calculated to determine its contribution to the target (Tao et al. 2023). The global contribution of features was quantified using the average absolute SHAP values. In addition, the classification mapping strategy was used to integrate feature importance, enabling the analysis of the contributions of the three categorical features (Wang et al. 2024). The SHAP value was calculated according to Eq. (5):

$$SHAP(x) = \sum_{S\subseteq M/\{j\}} \frac{(|S|-1)!(|M|-|S|-1)!}{|M|!}\left(f(x_s\cup\{j\}) - f(x_s)\right) \tag{5}$$

where $|M|$ is the size of the feature set, S is a subset of features in addition to feature j, $|S|$ is the size of the subset S and $f(x_s)$ is the predicted value of subset S.

2.5. Monomer screening and optimization

After the development of machine learning models, approximately 8000 potential aqueous monomers were retrieved from Pubchem for use in the fabrication of polyester membranes. Notably, the Morgan fingerprint was utilized to represent the molecules to reduce the computational load associated with the DFT calculations. The t-distributed stochastic neighbour embedding (t-SNE) algorithm was subsequently used to visualize the chemical space generated by the fragment representation method. Each point in the t-SNE plot represents an aqueous monomer, whereas the distance between any two points reflects the degree of similarity between their molecular structures (Feng et al. 2025; Huang et al. 2025). Five candidate aqueous monomers were identified as structurally similar to high-performance monomers either within or outside the dataset and were selected to fabricate the polyester membranes. Finally, the input features of the fabrication and operation parameters for polyester membrane fabrication were further identified using the machine learning model integrated with Bayesian optimization.

3. Results and discussion

3.1. Statistical analysis of datasets

Pearson correlation analysis was conducted to evaluate the linear relationships among the different DFT descriptors, and the aim was to minimize feature redundancy. Generally, a correlation coefficient of 1 or −1 represents a significant positive or negative linear correlation between the variables. As shown in Fig. 2a, 16 DFT descriptors, with mostly low correlation coefficients below 0.7, were selected to describe the topological structure, charge distribution, polarity, and reactivity of aqueous monomers. Since the charged area increases with increasing molecule volume, the molecular diameter has a high correlation coefficient, with Vs_+ and Vs_- values of 0.79 and 0.89, respectively. PSA, which is the area with an absolute electrostatic potential exceeding 10 kcal/mol, had a strong correlation coefficient (>0.8) with both the Vs_+ and molecular volume. These findings were in agreement with established domain knowledge and confirmed the accuracy of the DFT descriptors (El-Desouky et al. 2022; Jing et al. 2024).

The distributions of six typical fabrication parameters were extracted to assess the validity of the dataset. As shown in Fig. 2b, the concentration of the aqueous monomers substantially varied (0.05–10.00 wt %), whereas that of the organic monomers varied within a narrow range (0.1–0.8 wt%). Given that the organic monomer used in this study was exclusively 1,3,5-benzenetricarbonyl chloride, the observed wide distribution indicated that different aqueous monomers had distinct optimal concentrations for fabricating polyester membranes with specific performance requirements. To increase the reactivity of the hydroxyl functional groups on the aqueous monomers, the deacid reagent was introduced at concentrations predominantly ranging from 0.4 to 1.5 wt%. Compared with the fabrication process of polyamide membranes,

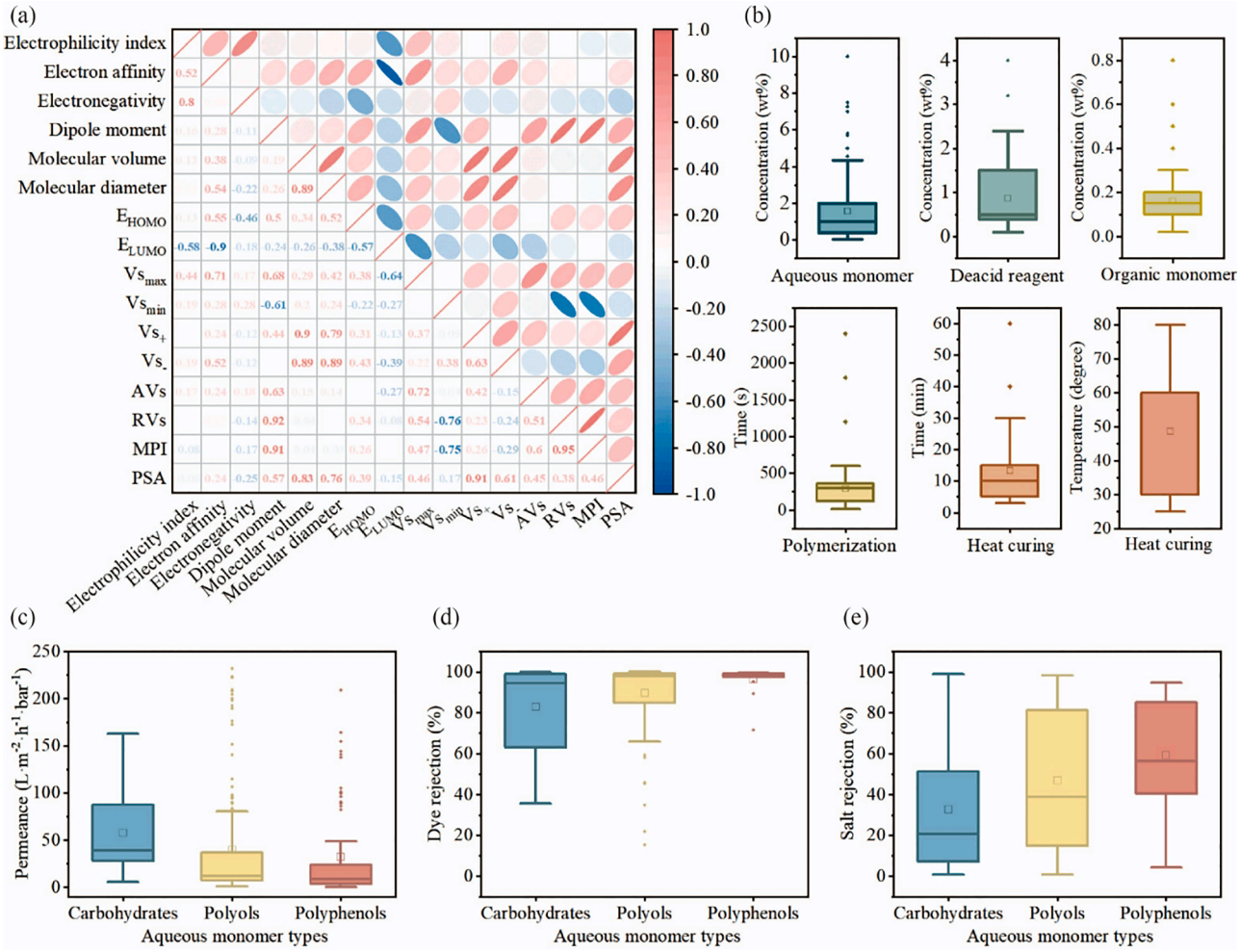


Fig. 2. (a) Pearson correlation for DFT descriptors. The elongated ellipse suggests a strong linear correlation between the variables, whereas an almost circular ellipse indicates a weaker correlation. The major axis inclined to the upper right signifies a positive correlation, while one inclined to the lower right denotes a negative correlation between the variables. (b) Box diagram of fabrication parameters in the dataset. Separation performance of polyester membranes fabricated by three types of aqueous monomers: (c) permeance, (d) dye rejection and (e) salt rejection.

an extended polymerization time was applied to increase the cross-linking degree of the polyester membranes, with a median value of 300 s (Jiang et al. 2025; Lau et al. 2015). For the heat curing parameters, the heat curing time was mainly distributed in the range of 3–80 min, with temperatures between 25 and 120 °C. In addition, the Pearson correlation coefficients between the DFT descriptors and fabrication parameters were all <0.5 (Fig. S2), indicating a low degree of linear correlation. Based on these results, further analysis was needed to elucidate the nonlinear relationships among the parameters involved in the fabrication process of the polyester membranes.

The data distribution of polyester membrane performance was statistically analysed statistically. Based on the molecular structure of the aqueous monomers, polyester membranes were fabricated from three categories of aqueous monomers: carbohydrates, polyols and polyphenols (Park et al. 2024; Wu et al. 2024). As shown in Fig. 2c, all polyester membranes had high permeance due to their loose structure and the low reactivity of the aqueous monomers. Compared with polyol-based and polyphenol-based membranes, carbohydrate-based membranes presented the highest permeance, ranging from 28.8 to 86.6 $L{\cdot}m^{-2}{\cdot}h^{-1}{\cdot}bar^{-1}$ within the 25 % to 75 % dataset. In addition, the polyol-based and polyphenol-based membranes had more outliers, and the maximum permeance was over 200 $L{\cdot}m^{-2}{\cdot}h^{-1}{\cdot}bar^{-1}$. These results indicated that the fabrication conditions also played an important role in determining the permeance of these membranes. With respect to dye rejection, the median values of carbohydrate-based and polyol-based membranes were 94.8 % and 97.6 % (Fig. 2d), respectively. In contrast, polyphenol-based membranes had a narrow distribution in terms of dye rejection, with a median value of 99.2 %. Moreover, all types of the polyester membranes showed wide distributions of salt rejection (Fig. 2e). Due to the simultaneous data distribution, including intervals of high dye rejection and low salt rejection, the polyester membrane has the potential for textile wastewater recovery after optimizing the fabrication conditions.

3.2. *Model construction and evaluation*

Machine learning is a promising approach for analysing and understanding the multidimensional parameters involved in polyester membrane fabrication (Hu et al. 2025; Pang et al. 2025; Tao et al. 2023). Before model development, numerical features were standardized using the power transformer to minimize the impact of skewed data distributions, whereas categorical features were converted into numerical variables using backward difference encoding. This transformation allowed the models to recognize the categorical features and reduce noise within the dataset. XGBoost and CNN models were subsequently developed, and their hyperparameters were optimized using Bayesian optimization to ensure the creation of reliable predictive models for polyester membrane performance. Moreover, 5-fold cross-validation

was used to avoid overfitting when model training and optimization were performed. When the trained model was obtained, the water permeance, dye rejection, and salt rejection were predicted for all data points within the testing set. The distributions of the actual and predicted values are displayed in joint scatter plots (Fig. 3). The oblique dashed line in the scatter plots showed perfect agreement between the predicted and actual values. Thus, tight clustering of data points around this line indicated better model prediction performance (Chen et al. 2025).

The generalization ability of the XGBoost and CNN models was further assessed on the unseen testing set using three evaluation metrics. As shown in Table 1, the XGBoost and CNN models achieved R^2 values of 0.85 and 0.77, respectively, for predicting the permeance of the polyester membranes. The higher R^2 value of the XGBoost model was attributed to the inclusion of a regularization term and the use of second derivative information in the loss function, this enabled the high adaptability to the low-dimensional and small-volume dataset with the hidden higher-order correlation relationships. Similar results also revealed that the XGBoost model exhibited greater generalization ability than artificial neural networks in predicting membrane performance when datasets containing hundreds to thousands of samples were used. The R^2 values of the XGBoost model were further improved to 0.88 and 0.87 for dye and salt rejection, respectively, because of the use of larger datasets for model training. In addition, the XGBoost model demonstrated lower *MAE* and *RMSE* values than the CNN model for predicting polyester membrane performance. The strong generalization ability of the XGBoost model showed the rationality of the input features and the effectiveness of dataset preprocessing. Compared with machine learning models assisted by the other molecular descriptors derived from the count descriptors, fingerprints and topological descriptors (Gao et al. 2022; Gong et al. 2024; Lee et al. 2024), the DFT-assisted machine learning model achieved better prediction performance for membranes (Table S9). This result indicated that DFT descriptors provided more detailed information on the molecular structures since they conformed to quantum principles. In summary, the XGBoost model outperformed the CNN model with higher R^2 values and lower *RMSE* and *MAE* values; thus, it was the preferred choice for further feature decoupling and importance analysis.

Table 1
Polyester membrane performance prediction by XGBoost and CNN models.

Model	Targets	R^2	*MAE*	*RMSE*
XGBoost	Permeance	0.85	11.35	20.39
	Dye rejection	0.88	5.90	5.80
	Salt rejection	0.87	7.59	11.71
CNN	Permeance	0.77	13.23	24.56
	Dye rejection	0.80	4.79	7.54
	Salt rejection	0.78	9.69	15.03

3.3. Feature importance assessment

The contributions of each input feature to polyester membrane performance were analysed using SHAP based on the developed XGBoost model. A positive Shapley value indicates that the input feature contributes positively to the target, whereas a negative Shapley value signifies a negative contribution (Li et al. 2022a; Xu et al. 2024). The top 20 features influencing the targets were extracted and presented in the SHAP summary plots (Fig. S3). As shown in Fig. 4a, the transmembrane pressure, aqueous monomer concentration, substrate pore size, Vs_{max} and electronegativity were identified as the five most influential parameters for predicting the membrane permeance. Notably, the operation parameter of transmembrane pressure had the highest importance value of 19.1 for membrane permeance. Theoretically, membrane permeance is defined as the membrane flux normalized by the transmembrane pressure; thus, it is independent of pressure variations. However, the high transmembrane pressure had a significantly negative effect on the membrane permeance. This negative effect of transmembrane pressure on permeance was attributed to the compaction of

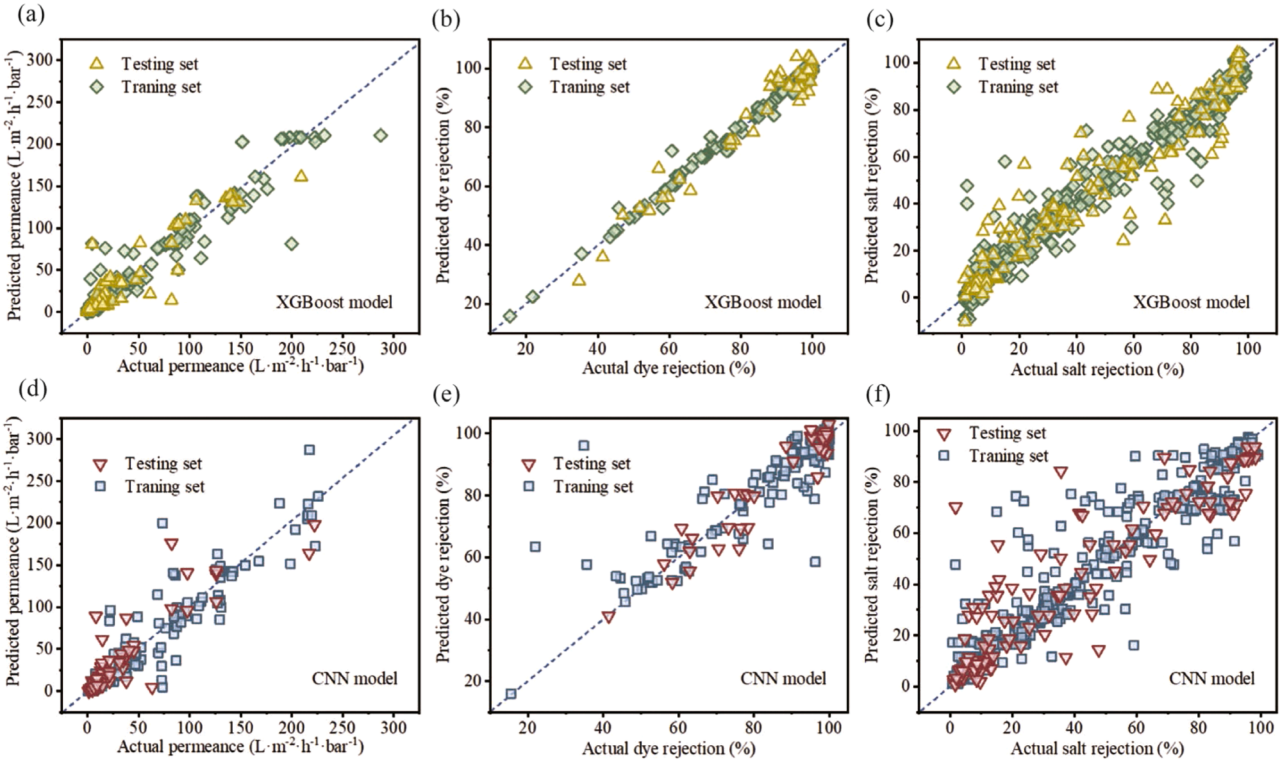


Fig. 3. Prediction results of (a) permeance, (b) dye rejection and (c) salt rejection by the XGBoost model. Prediction results of (d) permeance, (e) dye rejection and (f) salt rejection by the CNN model.

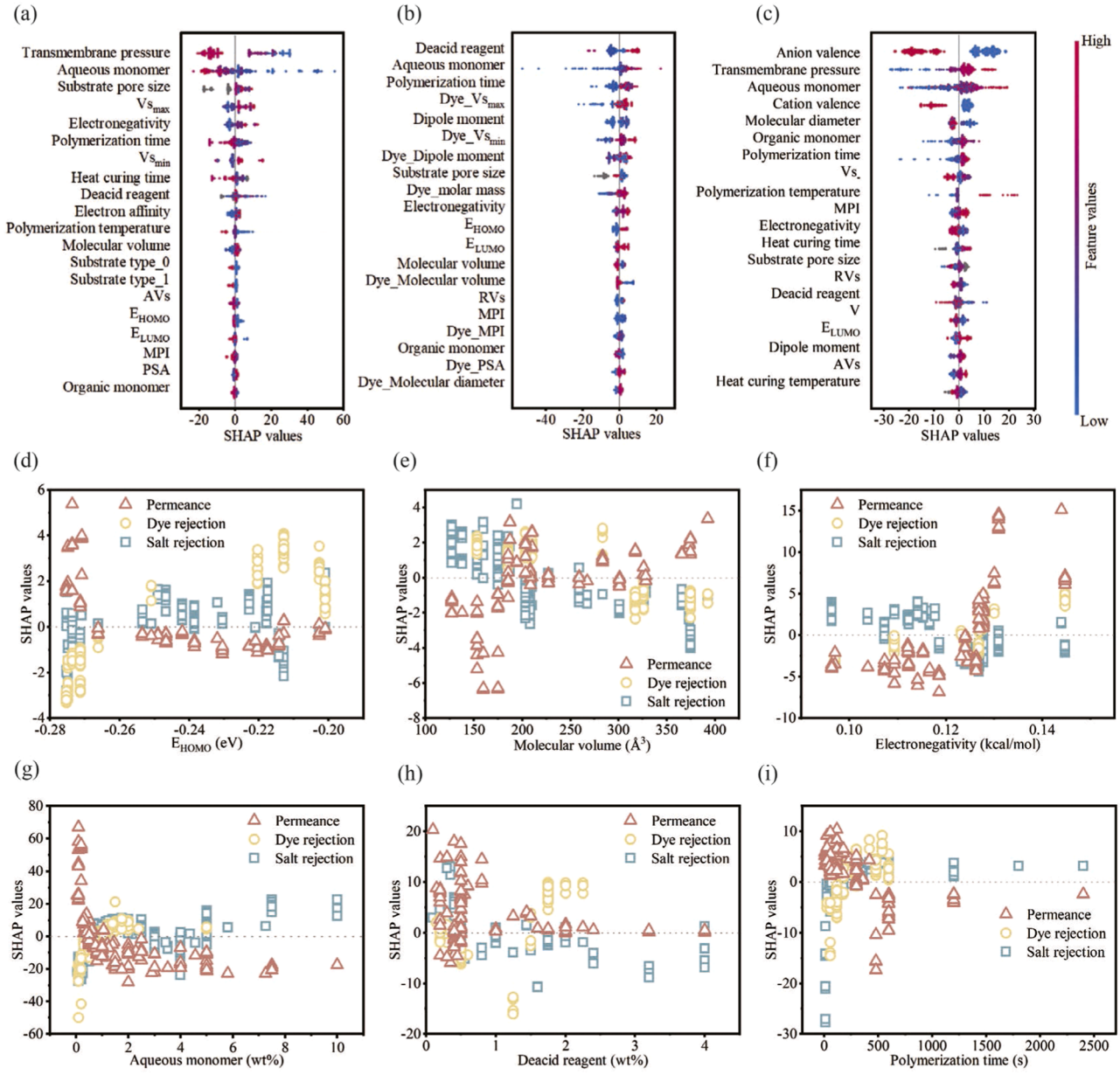


Fig. 4. SHAP summary plots of top 20 features for (a) permeance, (b) dye rejection and (c) salt rejection. Partial dependence plot of (d) E_{HOMO}, (e) molecular volume, (f) electronegativity, (g) aqueous monomer concentration, (h) deacid reagent concentration and (i) polymerization time.

the polyester layer and the substrate and resulted in a reduction in the pore size and porosity and an increase in the resistance of the polyester membrane (Ding et al. 2024). In addition, compared with those of the polyamide membranes, the loose structure of polyester membranes further amplified the effect of transmembrane pressure on compaction (Zhao et al., 2022b).

Dye rejection was influenced mainly by the deacid reagent concentration, aqueous monomer concentration and polymerization time (Fig. 4b). In addition, the dye properties, such as Vs_{max}, Vs_{min} and its diameter, significantly influenced its rejection. Since the polyester membranes generally had a negative charge because the carboxyl functional groups originated from the hydrolysis of acyl chloride groups, positively charged dyes tended to be absorbed into the polyester membrane, whereas negatively charged dyes were rejected through Donnan exclusion (Yao et al. 2024). The separation performance of polyester membranes was also governed by size sieving; here, the dyes with larger diameters were retained by the smaller pores and resulted in high removal efficiency (Yan et al. 2023; Yang et al. 2023). The positive correlation between dye diameter and removal efficiency aligned with this domain knowledge.

With respect to salt rejection, as shown in Fig. 4c, anion and cation valences were identified as important contributors to the polyester membranes. This was attributed to the transport of the salt through the polyester membranes predominantly governed by the Donnan exclusion, which repels co-ions while attracting counterions (Quezada-Renteria et al. 2024). The transmembrane pressure was also positively correlated with the salt rejection. Unlike the mechanism affecting membrane permeance, the high transmembrane pressure led to concentration polarization on the membrane surface, thereby alleviating the transport of salt through the membrane. Notably, the fabrication parameter of the aqueous monomer concentration had a high contribution to all targets. A high aqueous monomer concentration yielded a denser polyester layer, reducing the water permeance and enhancing the dye and salt rejection (Zhao et al. 2022b).

To further examine the impact of the key features on the polyester membrane performance during the fabrication process, three DFT

descriptors and three fabrication parameters were selected for the marginal effect analysis based on the SHAP analysis. Specifically, a higher E_{HOMO} is associated with greater dye and salt rejection but lower membrane permeance (Fig. 4d). Since E_{HOMO} is associated with the electron-donating ability of a molecule, aqueous monomers with a high E_{HOMO} are more likely to react with 1,3,5-benzenetricarbonyl chloride, resulting in the formation of denser polyester membranes (Wang et al. 2023). In contrast, the molecular volume exhibited an opposite trend in terms of the membrane performance with respect to that of E_{HOMO} (Fig. 4e). This was likely attributed to the aqueous monomers with high molecular volumes featuring complex three-dimensional topologies and multiple reactive sites (Yuan et al. 2020), potentially leading to a loose structure or defects in the polyester layer. Notably, the electronegativity of the aqueous monomers was significantly positively related to permeance and dye rejection, but negatively related to salt rejection (Fig. 4f). Due to its opposing effects on the dye and salt separation, aqueous monomers with high electronegativity could contribute to the fabrication of the polyester membranes with high permeance and improved dye/salt selectivity.

In terms of the fabrication parameters, the aqueous monomer concentration was negatively correlated with the permeance when it exceeded 0.5 wt% (Fig. 4g) but positively correlated with dye and salt rejection once it exceeded 0.6 wt% and 0.8 wt%, respectively. These findings indicated that an aqueous monomer concentration of 0.6–0.8 wt% could produce a membrane with high dye rejection but relatively low salt rejection. For the deacid reagent (Fig. 4h), no clear trend was observed at low concentrations (<1.0 wt%). These results were attributed to the poor deprotonation behaviour of the hydroxyl functional groups in aqueous monomers under weakly basic conditions. Above 1.0 wt%, the concentration of the deacid reagent was positively correlated with the dye rejection and negatively correlated with the salt rejection, whereas its effect on permeance remained negligible. Notably, an excessive concentration of the deacid reagent could lead to hydrolysis of the polyester layer, resulting in the formation of membrane defects (Liang et al. 2020). However, this phenomenon was not observed because of the absence of defective membrane performance in the dataset. The influence of the polymerization time on the performance of the polyester membrane was similar to that of the aqueous monomer concentration (Fig. 4i).

A quantitative comparison based on a classification mapping strategy was conducted to evaluate the contributions of the three input feature categories of the DFT descriptors, fabrication parameters and operation parameters to the performance of the polyester membranes. As shown in Fig. 5a, the fabrication parameters, particularly the aqueous monomer concentration and substrate pore size, exhibited a dominant contribution of 43.2 % to the membrane permeance. Even though the transmembrane pressure was the sole operation parameter influencing permeance, its significant contribution of up to 25.8 % revealed the critical role of enhancing the mechanical strength of the membrane in improving permeance. For dye rejection, the DFT descriptors had the highest contribution at 39.9 % (Fig. 5b), surpassing the contributions of the fabrication parameters (33.2 %) and operation parameters (26.9 %). In contrast to the contribution trend observed for dye rejection, the operation parameters accounted for the largest contribution (40.9 %) toward salt rejection because of the distinct impact of the ion valence on membrane separation. Since the dominant contributing factors for each performance target are different, the simultaneous optimization of DFT descriptors, fabrication parameters, and operation parameters is crucial for achieving high-performance polyester membrane applications.

3.4. Polyester membrane design based on machine learning for textile wastewater treatment

Fig. 6a shows the chemical spaces occupied by approximately 8000 potential aqueous monomers using the t-SNE algorithm, this algorithm projects high-dimensional molecular feature vectors into a low-dimensional space. In the newly constructed low-dimensional space, similar aqueous monomers are positioned closer together, whereas dissimilar monomers are placed farther apart. The aqueous monomers in the dataset are clearly uniformly distributed within the unexplored chemical space and are represented by the green circles in Fig. 6a. Based on the performance of polymer membranes in the dataset, five aqueous monomers from the dataset were selected for further optimization of fabrication and operation parameters using the XGBoost model combined with Bayesian optimization. In addition, five candidate aqueous monomers were identified in an unexplored chemical space, positioned near the previously selected aqueous monomers. The molecular formulas of ten aqueous monomers are displayed alongside the t-SNE plot. Three of the aqueous monomers are classified as carbohydrates, three as polyphenols, and the remaining four as polyols.

During Bayesian optimization, the water permeance was targeted at over 100 $L{\cdot}m^{-2}{\cdot}h^{-1}{\cdot}bar^{-1}$, with Congo red rejection exceeding 95 % and Na_2SO_4 rejection below 20 %. According to the predictions from the XGBoost model, the polyester membranes fabricated using ten different

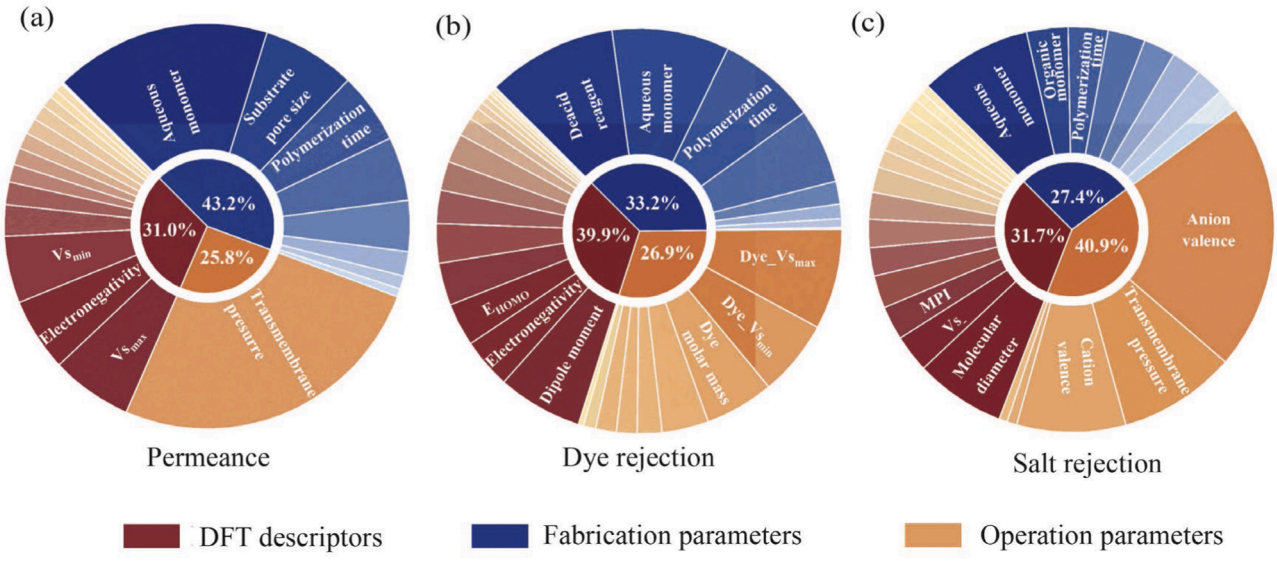


Fig. 5. Quantitative contribution of input features to the (a) permeance, (b) dye rejection and (c) salt rejection model. The input features were categorized into DFT descriptors, fabrication parameters, and operation parameters.

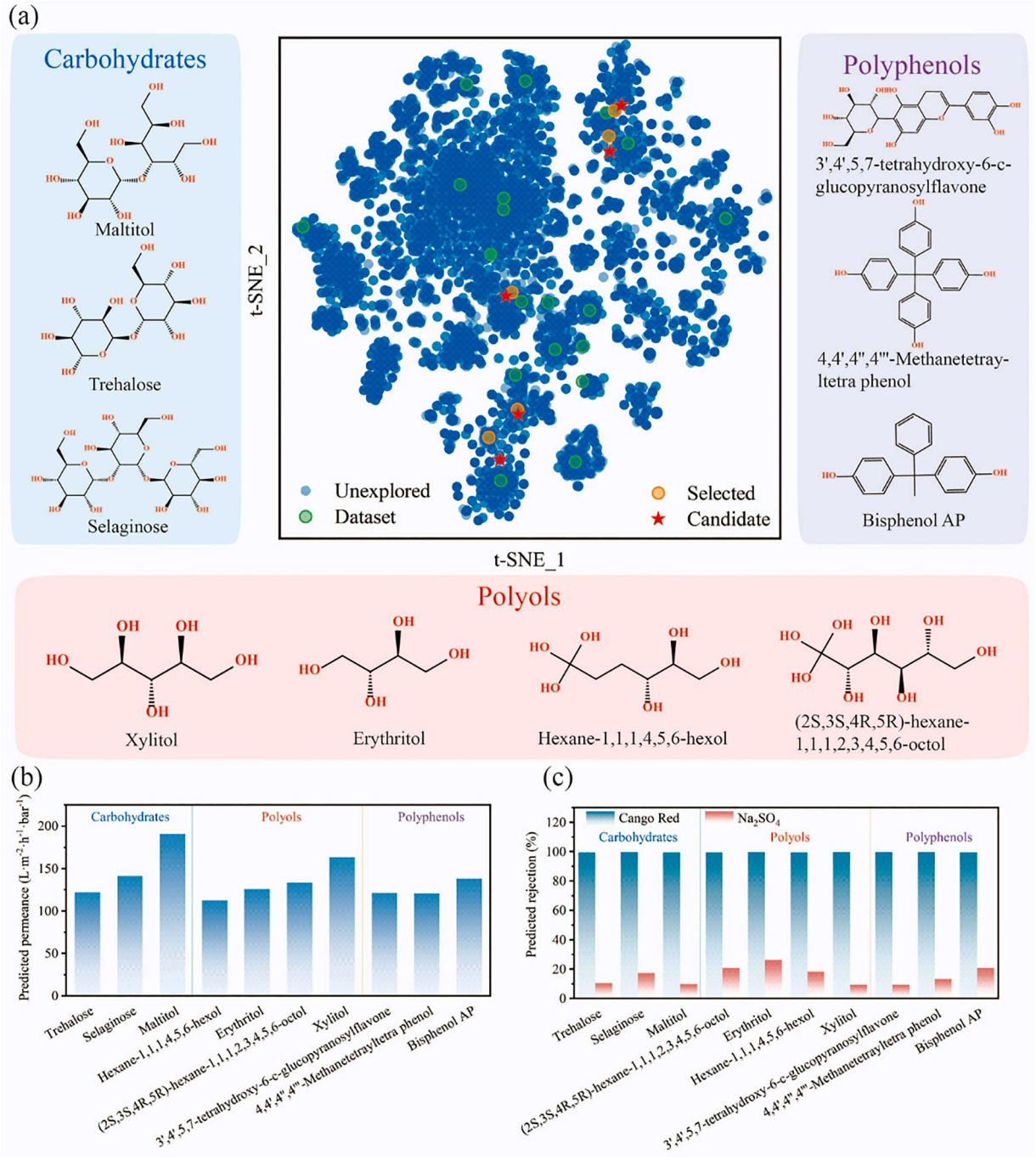


Fig. 6. (a) The t-SNE plot depicts the distribution of aqueous monomers within the chemical space. Green circles represent the aqueous monomers from the dataset, yellow circles indicate aqueous monomers for high-performance polyester membranes identified within the dataset, and five-pointed stars denote the candidate aqueous monomers. The molecular formulas of the selected and candidate aqueous monomers are displayed alongside the t-SNE plot. Predicted performance of (b) permeance and (c) rejection for polyester membranes fabricated using ten different aqueous monomers.

aqueous monomers all met these conditions (Fig. 6b and c), and their corresponding fabrication parameters are listed in Table S10. Xylitol was identified as the optimal candidate for fabricating membranes with the highest Congo red/Na_2SO_4 selectivity of 286 %, although its water permeance was lower than that achieved with 4,4′,4″,4‴-methanetetrayltetraphenol and bisphenol AP. Since xylitol belongs to the polyol family, its performance aligns with the statistical analysis (Fig. 2c–e) showing that polyols are more suitable for dye/salt selectivity. In addition, xylitol, which is outside the dataset, demonstrated that the developed model can identify aqueous monomers from unexplored chemical space, thereby enabling the efficient design of high-performance polyester membranes.

Dye/salt selectivity is an important step in textile wastewater recovery. To validate the fabrication process identified by the XGBoost model, five polyester membranes were fabricated, and their specific fabrication details are provided in Text S2 and Table S10. The similar values between predicated and test membranes further demonstrated the model could predicate the polyester membrane performance (Fig. 7a). The xylitol polyester membrane showed the best performance with a dense surface covering the pores of the substrate (Fig. S4). At pH 7, the zeta potential of the xylitol polyester membrane was −24.1 mV (Fig. S5), indicating that rejection was governed by size sieving rather

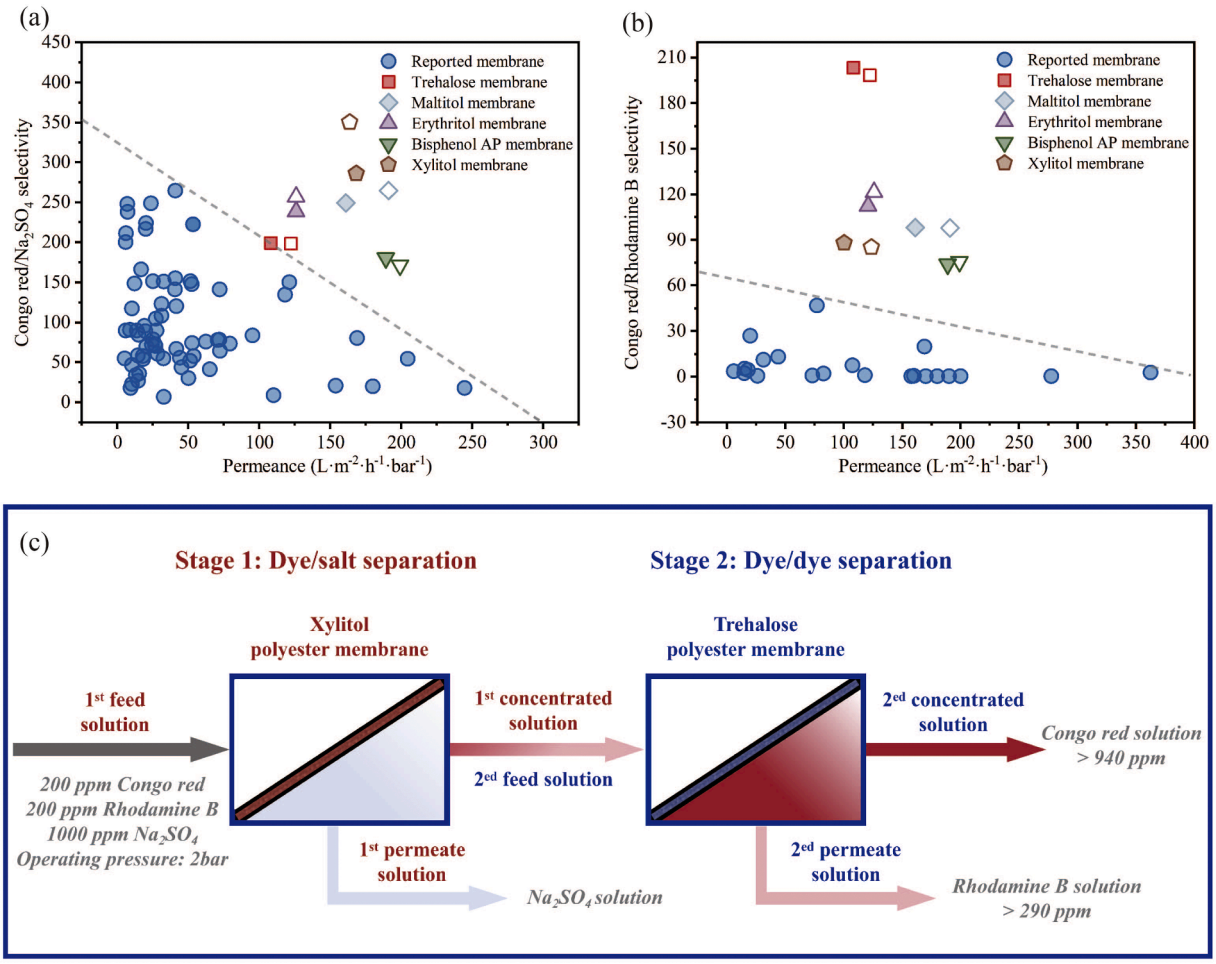


Fig. 7. (a) The trade-off between water permeance and Congo red/Na_2SO_4 selectivity of polyester membranes. (b) The trade-off between water permeance and Congo red/rhodamine B selectivity of polyester membranes. The hollow and solid legends are predicted and test values, respectively. (c) Schematic diagram of a two-stage membrane process for textile wastewater recovery utilizing polyester membranes fabricated by xylitol and trehalose. The model textile wastewater contained 200 ppm Congo red, 200 ppm rhodamine B and 1000 ppm Na_2SO_4.

than the Donnan exclusion. Since salt rejection primarily depends on the Donnan exclusion, reducing the zeta potential facilitates the permeation of the salt ions through the membrane. Notably, the molecular weight cut-off of the xylitol polyester membrane was 1840 Da (Fig. S6). This value exceeded the molecular weights of both Congo red and rhodamine B. However, the high dye rejection performance indicated that dye molecule aggregation occurred in aqueous solution at high salt concentrations (Fig. S7), likely caused by the compression of the electrical double layer (Sriram et al. 2024). Generally, the dye/salt selectivity exhibited an inverse relationship with water permeance, which is commonly referred to as the trade-off effect (Jin et al. 2022). Using Congo red and Na_2SO_4 as representative compounds for dye and salt selective rejection, the xylitol polyester membrane achieved both high permeance and exceptional dye/salt selectivity (Fig. 7a) to successfully overcome the trade-off effect. To fully leverage the dataset on dye and salt rejection, the XGBoost model independently predicted the dye and salt rejection, which could introduce errors in the estimation of dye/salt selectivity. Nevertheless, the comparable performance between the predicted and experimental results indicated that the XGBoost model exhibited an acceptable error margin in predicting the dye/salt selectivity of polyester membranes.

Another important step in textile wastewater recovery is the separation of dye mixtures at the molecular level. Following a similar approach to fabricate a high-performance dye/salt selectivity membrane, trehalose was observed as the optimized aqueous monomer for the fabrication of a polyester membrane designed for dye/dye selectivity (Fig. 7b). Compared with the xylitol polyester membrane, the trehalose polyester membrane had greater surface electronegativity (Fig. S5) and a narrower pore size distribution (Figs. S6, S8). These properties were advantageous for the selective separation of dyes with different charges because negatively charged dyes are readily retained through the Donnan exclusion, whereas neutrally charged dyes can permeate the negatively charged membrane (Zhao et al. 2023).

Resource recovery from textile wastewater not only reduces treatment challenges but also facilitates the conversion of waste into valuable resources. To meet this goal, a two-stage membrane process was developed to treat textile wastewater for effective resource recovery. This process consists of two main stages: the separation of dyes from salts, followed by the purification of the individual dyes from the resulting mixture. The performance of the process was evaluated using a model textile wastewater containing Congo red, rhodamine B and Na_2SO_4 as the feed solution (Fig. 7c). In the first stage, the xylitol polyester membrane was used for dye/salt selectivity. To achieve high permeation of Na_2SO_4, the rejection rate of rhodamine B was intentionally maintained at 73.6 %, whereas that of Congo red was 99.7 %. In the second stage, the concentrated solution from the first stage was further separated by a trehalose polyester membrane. The filtration experiment revealed that Congo red was predominantly retained in the second concentrated solution, whereas rhodamine B was transported through the membrane. Based on these data, Congo red and rhodamine

B were successfully recovered from the concentrated and permeate solutions, respectively, demonstrating the feasibility of the two-stage membrane process using machine learning-guided polyester membranes. Since the permeance of polyester membranes was ten times greater than that of traditional nanofiltration membranes, the multi-stage membrane separation process could further increase the purity of the dye with high energy efficiency. Although this work used only model textile wastewater, the XGBoost model has the potential to design the required membranes tailored to the characteristics of practical textile wastewater. In addition, other types of aqueous monomers can be incorporated into DFT-assisted machine learning models to develop high-performance nanofiltration membranes for textile wastewater recovery applications.

4. Conclusions

In summary, DFT-assisted machine learning models were proposed for polyester membrane design in textile wastewater recovery applications based on the input features from three categories: DFT descriptors, fabrication parameters and operation parameters. A precise and robust XGBoost model was developed to reveal the nonlinear relationships embedded within the dataset and accurately predict the performance of polyester membranes. Using SHAP analysis, both local and global contributions of input features were identified, and the key factors during the fabrication process that influence the polyester membrane permeance and the dye and salt rejection were highlighted. In addition, aqueous monomers were screened from an unexplored chemical space of 8000 candidates using the developed model combined with Bayesian optimization. Five polyester membranes were then fabricated using the identified parameters, and the dye/salt and dye/dye selectivities with upper bound-breaking performance were obtained. The two membranes were employed in a two-stage membrane process to enable the efficient recovery of the dyes from textile wastewater.

CRediT authorship contribution statement

Peng Liu: Writing – original draft, Investigation. **Hangbin Xu:** Investigation. **Pengrui Jin:** Validation, Investigation. **Xuewu Zhu:** Validation, Investigation. **Junfeng Zheng:** Formal analysis. **Yanling Liu:** Visualization, Investigation. **Jiaxuan Yang:** Visualization. **Daliang Xu:** Writing – review & editing, Conceptualization. **Heng Liang:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The work was jointly supported by the National Natural Science Foundation of China (No. 52300083), China Postdoctoral Science Foundation (No. 2024T171148), the special fund of State Key Joint Laboratory of Environment Simulation and Pollution Control (No. 24K22ESPCT), the Jinan City School Integration Development Strategy Project (No. JNSX2024032), Open Project of State Key Laboratory of Urban Water Resource and Environment, Harbin Institute of Technology (No. QA202323).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.watres.2025.123438.

Data availability

Data will be made available on request.

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